

Faculty of Information Technology**Semester 2, 2026****FIT5145: Foundations of Data Science****Assignment 2: Description****Due Date: Friday, Week 8 (Sep 18, 2026), 11:55 PM**

The aim of this assignment is to investigate and visualise data using **R**. It will test your ability to:

1. Read data files and extract related data from those files;
2. Clean and process data into the required formats;
3. Perform exploratory data analysis and visualisation;
4. Use basic tools for managing and processing data; and
5. Communicate your findings in your report.

Assessment Details:

- Assessment Type: Individual Assignment
- Total marks: 15%
- Due Date: Friday, Week 8 (Sep 18, 2026), 11:55 PM. Please notice that we do not accept submissions after Sep 25, 2026 (i.e., 7 days after the due date).

Submission Details:

You are required to submit two separate files: a **PDF report** and an **RMD file**.

1. **PDF report:** Please convert (knit) your RMD file to HTML, then print the HTML file using the “Save as PDF” option under Destination. Your report must include *(a) code, (b) code output, and (c) an explanation* for each question.

(a) Code: For better readability, please use separate code chunks for each question or task.

(b) Code output: Please ensure that the outputs from your code and any written answers are clearly included for each question.

(c) Explanation: Please explain how you answered each question (i.e., explaining your codes or summarising your work for each question).
2. **RMD file:** This is the original R Markdown (RMD) file that you used to generate the PDF report. Please include both your code and explanations in this file.

Notes:

1. Whenever you identify anomalies, errors, or any issues in the dataset that may negatively affect the validity of your analysis, **you are expected to determine and justify an appropriate approach and perform the necessary data wrangling.**

This assignment also assesses your ability to critically evaluate data quality and make informed preprocessing decisions. Different approaches may be acceptable, provided they are clearly justified and appropriate for the analysis.

2. Whenever a question asks for a certain value, your code should produce the value. For example, when a question asks for the number of rows contained in a table, your code should print out the number of rows. **Extraction of the answer manually by eye-examination will not earn any marks.**
3. **Please check whether you can highlight texts in your PDF**, as shown below. This ensures the Turnitin score can be generated properly for your PDF file (we only need the Turnitin score for the PDF, not the RMD file).

 **MONASH University**

Faculty of Information Technology
Semester 2, 2026
FIT5145: Foundations of Data Science
Assignment 2: Description
Due Date: Friday, Week 8 (Sep 18, 2026), 11:55 PM
The aim of this assignment is to investigate and visualise data using R. It will test your ability to:

1. Read data files and extract related data from those files;
2. Clean and process data into the required formats;
3. Perform exploratory data analysis and visualisation;
4. Use basic tools for managing and processing data; and
5. Communicate your findings in your report.

Assessment Details:

- Assessment Type: Individual Assignment
- Total marks: 15%
- Due Date: Friday, Week 8 (Sep 18, 2026), 11:55 PM. Please notice that we do not accept submissions after Sep 25, 2026 (i.e., 7 days after the due date).

Assignment Task:

As a data scientist at the Department of Climate Change, Energy, the Environment and Water (DCCEEW), my manager has asked me to analyse the [National Waste Report 2022](#) dataset to extract insights that can inform departmental policy, regulation, and investment decisions in the waste management sector. The dataset contains data on Australia's waste generation, recovery and fate for all waste streams and various material categories, covering the period from 2005 to 2021 (further description available [here](#)).

There are two data files, *Wastes.csv* and *Year_State_ID.csv* with their details provided below. A more description of the dataset is available [here](#). **These files have been modified for the task. Please download them from the Assessments page on Moodle.** My manager wants me to explore the data through the following sections, including data cleaning, exploration, and visualisation, to identify findings relevant to these policy, regulatory, and investment decisions.

- ***Wastes.csv***

Column	Description
Case_ID	ID of each waste process case.
Year_State_ID	ID of Year and State.
Category	A broad classification of waste material.
Type	A more detailed classification of waste material. For example, the category 'Metals' may be split into : 'Aluminium', 'Non-ferrous metals (ex. aluminium)', and 'Iron and steel'.
Stream	Describes the source of waste, comprising three options: municipal solid waste (MSW) from households and council operations; commercial and industrial (C&I) waste; and construction and demolition (C&D) waste.
Fate	The ultimate destination of the waste, comprising five options: disposal; recycling; energy recovery; long-term storage; and waste reuse.
Tonnes	The quantity of waste.
Core_Non-core	Core waste: Waste generally managed by the waste and resource recovery sector; comprises solid non-hazardous waste, hazardous waste (including liquids) and biosolids. Non-Core waste: It includes material from mining, minerals processing, agriculture and fishing.
Description	• Describes each waste process case and provides a score and feedback for each case. • Environmental Impact Score: Measures how much a waste process affects the environment, either positively or negatively, using a 10-point scale. - 10: Very positive impact - 1: Very negative impact

- ***Year_State_ID.csv***

Column	Description
ID	ID of Year and State
Year	Financial year. Data is presented for each year between 2006-2007 and 2020-2021

State	State or territory in which the waste was generated.
Economic_Growth	The increase in the production of total goods and services (e.g.: 3.07: 3.07 % increase, -1.31: -1.31% increase)

Let's take a look at the file *Wastes.csv*.

1. How many unique "*Category*" values are there in the data file (6 marks)?
2. How many negative feedback comments are in the 'Description' column with an environmental impact score of 2 or 3 (4 marks)?
3. For each *Category* value, write code to calculate the fractions of waste tonnes of different waste sources, and then draw a chart to visualise the fraction numbers specific to the *Category* value (4 marks).
4. Please add the 'Year' and 'State' values from *Year_State_ID.csv* to *Wastes.csv*, compute the total waste tonnes for each year and state, and store the result in a dataset named 'temp'. Then, use a single R function to compute and display the statistical information (i.e., Min, Max, and Mean) of the total waste tonnes for each state in 'temp' (Note: You may use multiple functions/commands to prepare the pre-processed data table, but when you compute and display the statistical information, you need to use a single R function.) (4 marks).
5. A manager wants to see a chart showing a yearly trend of total waste tonnes of food organics for each state. To draw the chart, please convert the Year-Year formats of all *Year* values into Year formats (4 marks) (**Note: Use the dataset with the converted year format for all subsequent questions.**)
6. Write code to display the most recycled waste *Type* and the most energy-recovered waste *Type* with the corresponding year. Write code to display the most increased waste *Type* over years (5 marks).
7. Please investigate the factors influencing environmental impact scores. (Note: You may use variables from both *Wastes.csv* and *Year_State_ID.csv* for your analysis.) Marks will be awarded based on the depth and breadth of your investigation, including analyses and discussions, as well as the strength of your reasoning in drawing insights and conclusions (5 marks).
8. A manager is deciding where to invest in large-scale tyre processing capacity, and she wants to understand the disposal size variation and disposal frequency for each state. Create a relevant plot to support the manager's decision. Justify your choice, and explain what the plot tells her. (5 marks)

(Note: When considering tyre disposal size variation, use a tonne range rather than the raw Tonnes values. To do this, create a new column, *Tonnes_range*, that assigns each record to one of your ranges (bins). Decide your own tonne ranges (e.g. 1000-2000, 2000-3000) and create at least 4 ranges.)

9. Using both original datasets and external datasets, please investigate the factors influencing the yearly trend of total C&D waste tonnes. Marks will be awarded based on the depth and breadth of your investigation, including analyses and discussions, as well as the strength of your reasoning in drawing insights and conclusions. (Note: Please provide the source links for any external dataset used) (5 marks).